ReCoMER: Reliability and Context Modeling to Mitigate Biases in Multimodal Conversational Emotion Recognition

Anonymous CVPR submission

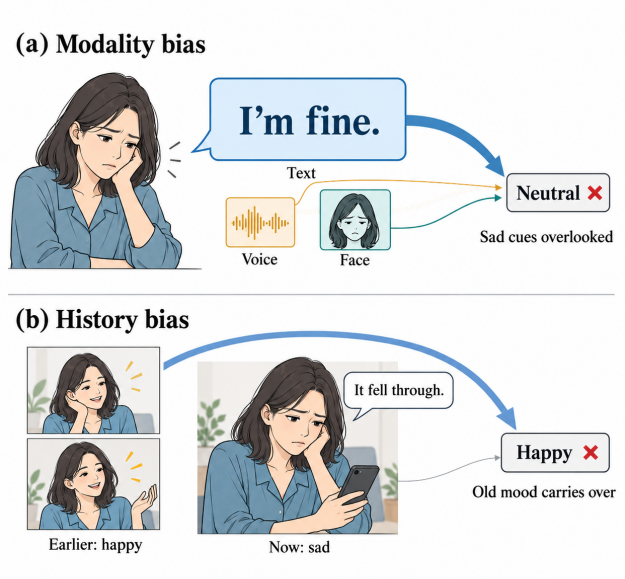
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Abstract

Conversational emotion recognition identifies the emotional state of each utterance in a dialogue from textual, acoustic and visual cues. However, models can exhibit history bias and modality bias when combining these cues with dialogue history. History bias occurs when outdated emotional context distorts judgments about the current utterance, whereas modality bias arises when overreliance on one modality obscures useful evidence from others. Thus, we propose ReCoMER, a unified framework that coordinates historical influence and modality contribution. To mitigate the history bias, we introduce Null-Candidate History Abstention (NCHA) with Feedback-Conditioned Context Reassessment (FCCR). NCHA uses a first-pass utility estimate to jointly guide second-pass competition with a no-history alternative and a shared admission gate, enabling history attenuation or rejection. FCCR learns modality-specific corrections through low-rank adapters, with norms bounded relative to current representations. To address the modality bias, Shapley-Anchored Bounded Evidence Routing (SABER) learns post-update modality contributions from Shapley-weighted marginal utilities under fixed historical conditions, enabling contribution-aware routing without subset evaluation at inference. At the prediction level, cRBEF integrates complementary audio-visual expertise through prior-referenced, gated posterior refinement and arithmetic fusion with the contextual prediction. Experiments on M3ED, IEMOCAP and EmotionTalk show ReCoMER outperforms state-of-the-art methods.

1. Introduction

Multimodal conversational emotion recognition identifies the emotion of each utterance from text, audio, visual cues and dialogue context. Tone and facial expression provide cues beyond the words, while preceding turns help interpret the current utterance. These capabilities matter for empathetic dialogue systems, intelligent tutoring, and human–robot interaction [21, 34].



Illustrative examples

Figure 1. Illustration of modality and history biases in conversational emotion recognition. (a) Modality bias: overreliance on the words "I'm fine" can lead to a neutral prediction, overlooking sadness conveyed by the voice and facial expression. (b) History bias: earlier happiness can obscure sadness in the current utterance.

We consider two related forms of biased evidence use: modality bias and history bias. Modality bias arises when one modality dominates despite useful cues in the others. In the illustrative example in Fig. 1(a), reassuring words accompany a strained voice and tense expression. Relying mainly on the words could produce a reassuring or neutral interpretation while missing the speaker's distress. History bias arises when earlier information receives too much influence over the current evidence. In Fig. 1(b), a speaker who was happy has become sad; carrying the earlier emotion forward could obscure this change. History can thus be relevant to the conversation without being helpful for the present emotion decision. These two biases are connected

because historical updates can change the representations from which modality contributions are estimated.

Existing methods have made partial progress on both problems. For history bias, HiRoC [19] combines relevance-weighted aggregation with sparse routing. Its history-enhancement step adds weighted historical features to the current representation. However, this step provides neither an explicit option to reject the combined history nor a constraint on its magnitude relative to the current representation. This matters because relevance alone does not ensure that history remains useful for recognizing the current emotion. Unhelpful history can still enter the representation; if its contribution is large, it may obscure cues of an emotional change and bias the prediction toward a previous emotional state. For modality bias, EMOE [7] dynamically adjusts modality weights for each sample while preserving the predictive ability of individual modalities. However, part of its weight supervision comes from how accurately each modality predicts on its own. This does not directly measure whether adding that modality improves the combined prediction. A modality that is less accurate alone may still help another modality distinguish otherwise ambiguous emotions. Giving such evidence too little weight can limit the benefit of combining modalities. Wei et al. [33] use sample-level Shapley values to estimate modality contributions and strengthen low-contributing modalities through training-time resampling. However, these estimates guide which modalities receive more training, rather than directly supervising how much the model should rely on each modality for a particular prediction. This weighting decision matters when different modalities provide conflicting emotion cues. Giving too much weight to misleading cues can cause the model to overlook evidence supporting the correct emotion.

To address history bias and modality bias, we propose ReCoMER, which coordinates historical admission, bounded feedback and contribution-aware modality allocation. Null-Candidate History Abstention (NCHA) first assesses historical utility through an initial attention pass, using current–history agreement, conflict and attention statistics. The resulting estimate adjusts competition between real history and a zero-valued null candidate in a second pass. A shared admission gate controls whether the resulting historical feedback is applied across modalities. Feedback-Conditioned Context Reassessment (FCCR) then transforms admitted historical summaries through modality-specific, scale-aware low-rank adapters and learned feedback scales. It bounds each correction relative to the corresponding current representation before applying the update. Shapley-Anchored Bounded Evidence Routing (SABER) estimates modality contributions from these updated representations. During training, it compares modality subsets under fixed historical admission and representations, measuring marginal changes in the correct-class log probability. The resulting Shapley targets supervise a scorer that produces bounded routing weights at inference without evaluating modality subsets. This sequence allows modality allocation to reflect the evidence produced by controlled historical feedback. While SABER regulates evidence within the contextual predictor, cRBEF provides a complementary prediction-level route for nonverbal information. It refines a separate text–audio–visual prediction using gated, prior-referenced evidence from an independently trained audio–visual expert. The corrected distribution is then combined with the contextual output through arithmetic fusion. We evaluate ReCoMER on M3ED, IEMOCAP, and EmotionTalk under dataset-specific configurations to validate this approach.

Our contributions are:

- We propose NCHA to condition second-pass null competition and a shared admission gate on an initial assessment of historical utility. This allows redundant history to be rejected before historical feedback is applied. FCCR then transforms admitted history into modality-specific learned corrections whose norms are bounded relative to the current representations.
- We develop SABER to supervise modality allocation with Shapley-based marginal utility targets computed after historical feedback. Modality subsets are compared under fixed historical conditions during training, while a learned scorer produces bounded routing weights without subset evaluation at inference.
- We introduce cRBEF to refine text–audio–visual predictions using gated, prior-referenced evidence from a separately trained audio–visual expert. Arithmetic fusion combines the refined distribution with the contextual output.

2. Related Work

2.1. Multimodal Emotion Recognition

Contextual methods model relations among utterances, speakers and modalities to support emotion prediction. DialogueRNN [21] tracks speaker states, while DialogueGCN [9] propagates information over utterance relations. MMGCN [15] and M3Net [4] extend graph modeling to multimodal interactions, and MultiEMO [27] captures cross-modal correlations through attention. HiRoC [19] further improves historical selection through relevance-weighted enhancement and sparse routing. These approaches enrich the representation or selection of contextual evidence. However, contextual relevance does not ensure that historical emotional cues remain useful when a speaker's emotion changes. Relevant history can still support an earlier emotional state and obscure the current one. In particular, HiRoC's [19] history-enhancement step adds aggregated history without an explicit rejection option

or a magnitude bound relative to the current representation. Addressing this problem requires distinguishing which history is relevant from whether its combined evidence should be used and how strongly it should affect current cues. ReCoMER makes these latter decisions explicit through NCHA with FCCR. NCHA uses an initial assessment of historical utility to regulate second-pass null competition and a shared admission gate, allowing historical feedback to be rejected. When history is admitted, FCCR transforms it into modality-specific, scale-aware learned corrections and bounds their norms relative to current representations. Together, these mechanisms control both the application and the strength of historical feedback, reducing the risk that earlier emotions override current evidence.

2.2. Modality Balance and Contribution Estimation

Methods for modality imbalance operate through training adjustment or sample-specific fusion. OGM-GE [23] modulates gradients to balance modality learning, while PMR [6] uses class prototypes to strengthen insufficiently learned modalities. Wei et al. [33] estimate sample-level Shapley contributions and use them to guide targeted resampling. These methods improve modality learning through optimization or training exposure. However, these balancing signals do not directly supervise modality weights for individual predictions. A modality may therefore learn useful cues that remain underused during fusion. Sample-specific fusion addresses this weighting decision explicitly. EMOE [7] learns dynamic modality weights and preserves unimodal predictive ability through distillation. Nevertheless, part of its routing supervision is derived from standalone prediction errors. A modality that performs poorly alone may still help another modality distinguish ambiguous emotions, so standalone accuracy does not directly measure its benefit to fusion. Underweighting such a modality can leave useful complementary evidence underused. This motivates supervising fusion weights with the predictive improvement obtained by adding each modality to the others. In ReCoMER, historical feedback changes the representations being combined, so this improvement is evaluated after feedback. SABER compares modality subsets under fixed historical admission and updated representations, using Shapley-weighted changes in correct-class log probability as routing targets. These targets train a scorer to produce bounded weights, connecting each modality's contribution to its influence on prediction without subset evaluation at inference.

2.3. Prediction-Level Correction

Prediction-level correction allows nonverbal evidence to refine multimodal decisions. HiRoC [19] uses disagreement-gated audio and visual residuals with capped correction strength. However, these learned residuals need not vanish when nonverbal posteriors match the training class prior. cRBEF instead defines corrective evidence as the log ratio between a separately trained audio–visual expert's posterior and that prior. The correction therefore vanishes when the posterior matches the prior. A sample-dependent gate regulates its strength when refining a separate text–audio–visual prediction, and arithmetic fusion combines the corrected distribution with the contextual output.

3. Method

3.1. Task formulation and overview

For utterance t, ReCoMER predicts an emotion label $y_t \in \{1, \dots, C\}$ from text, audio, vision and preceding dialogue context. We index modalities by $\mathcal{M} = \{T, A, V\}$. A modality-specific projection E_m maps an input feature $x_t^m \in \mathbb{R}^{d_m}$ to a current token $z_t^m = E_m(x_t^m) \in \mathbb{R}^d$. The same projection encodes historical utterance j as h_j^m. Here d_m and d are the input and shared dimensions; availability masks identify observed modalities and valid historical positions.

ReCoMER coordinates historical influence with modality contribution (Fig. 2). Null-Candidate History Abstention (NCHA) conditions a second history reading on an admission estimate from a preliminary reading. Feedback-Conditioned Context Reassessment (FCCR) applies the resulting history as a modality-specific, norm-bounded residual. Shapley-Anchored Bounded Evidence Routing (SABER) estimates contributions from the updated representations under a fixed historical state. Separately, cRBEF uses a text-free AV expert to correct a TAV prediction, which is arithmetically fused with the contextual output.

3.2. History admission and bounded feedback

NCHA decides whether to admit a history-dependent update; FCCR limits how strongly it changes current evidence. Together, they retain a direct current-token path while controlling historical refinement.

NCHA: conditional null competition. NCHA distinguishes historical relevance from admission through two attention passes. The first characterizes current–history agreement; its admission estimate conditions the second. For modality m, let $\mathcal{H}_t^m$ contain the valid preceding positions. Learned relative-position and speaker-relation embeddings augment h_j^m to form a key $\kappa_{t,j}^m \in \mathbb{R}^d$. A learned null key $n_m \in \mathbb{R}^d$ supplies an alternative whose *value vector is zero*. For attention pass $k \in \{1, 2\}$, we compute

$$\begin{aligned} s_{t,j}^{m,(k)} &= Q_m(z_t^m)^\top \kappa_{t,j}^m / \sqrt{d}, \\ s_{t,\varnothing}^{m,(k)} &= Q_m(z_t^m)^\top n_m / \sqrt{d} + b_t^{(k)}, \end{aligned} \tag{1}$$

where Q_m is an affine query projection. The first pass uses $b_t^{(1)} = 0$ and provides label-free statistics ξ_t:

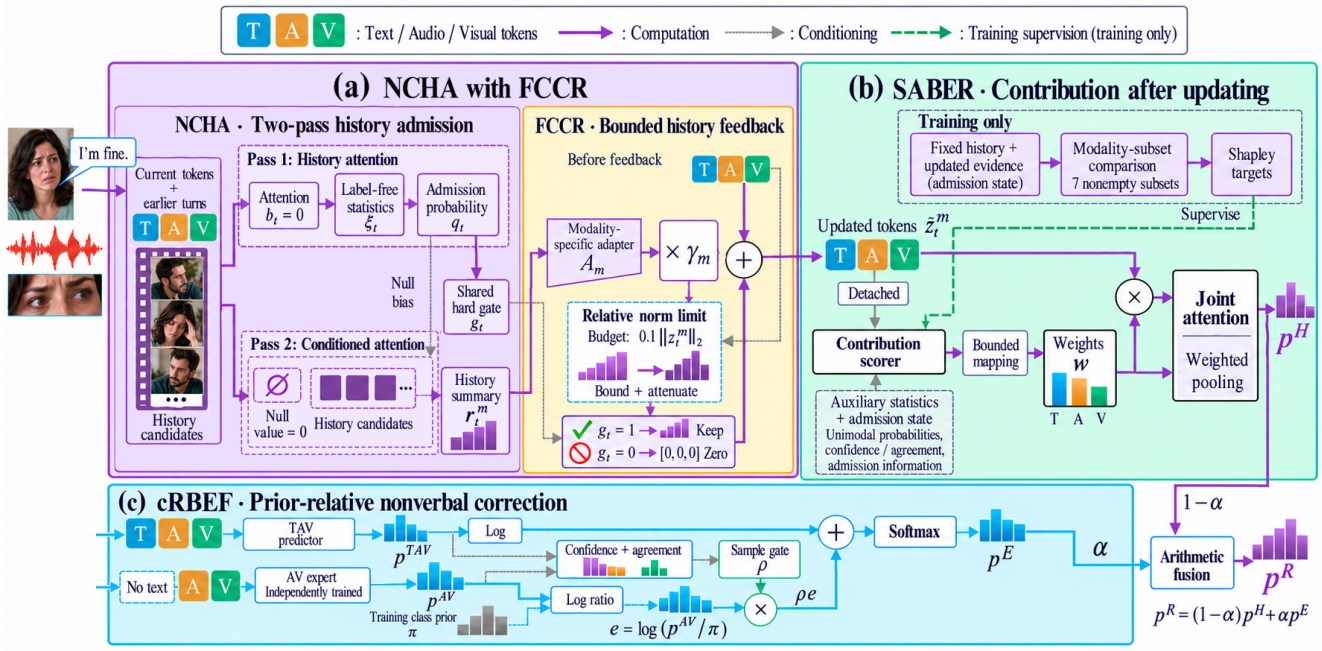


Figure 2. ReCoMER's flow. (a) NCHA conditions null competition and shared admission; FCCR bounds the current-token update. (b) SABER routes updated evidence using fixed-state Shapley supervision during training. (c) cRBEF applies gated, prior-relative AV correction and arithmetic fusion. Repeated g_t and w_t denote shared controls. The displayed budget instantiates η in Eq. (5).

confidence, current–history conflict, attention entropy, and speaker/recency contrasts. An MLP f_h gives

$$q_t = \sigma(f_h(\xi_t)/\tau_h), \qquad b_t^{(2)} = \beta \log \frac{1 - q_t}{q_t}. \tag{2}$$

Here σ is the sigmoid, $\tau_h > 0$ is a temperature, and $\beta \geq 0$ controls the null adjustment. Numerically stabilized logarithms are used. For $\beta > 0$, a smaller q_t increases the second-pass null score, reducing the attention mass assigned to real history.

The second pass normalizes real and null scores jointly:

$$a_t^{m,(2)} = \mathrm{softmax}\left([s_{t,j}^{m,(2)}]_{j \in \mathcal{H}_t^m}, s_{t,\varnothing}^{m,(2)}\right), \qquad r_t^m = \sum_{j \in \mathcal{H}_t^m} a_{t,j}^{m,(2)} h_j^m. \tag{3}$$

The zero-valued null attenuates r_t^m through the attention denominator. A separate, shared inference decision is

$$g_t = \mathbb{I}[q_t \geq \theta_h]\,\mathbb{I}[\text{valid history exists}], \tag{4}$$

where θ_h is the admission threshold and $\mathbb{I}$ denotes an indicator. Null competition modulates historical mass; g_t can suppress the entire feedback update. Both controls derive from the same admission estimate.

FCCR: modality-specific bounded feedback. FCCR maps r_t^m through a modality-specific low-rank residual adapter A_m, which normalizes and rescales its input, and a learned scalar γ_m. Norm-controlled residual adaptation [39, 40] is applied here to dialogue history under NCHA admission, *before* contribution estimation:

$$v_t^m = \gamma_m A_m(r_t^m), \qquad \bar{v}_t^m = v_t^m \min\left(1, \frac{\eta \|z_t^m\|_2}{\max(\|v_t^m\|_2, \epsilon)}\right), \qquad \widetilde{z}_t^m = z_t^m + g_t \zeta_t^m \bar{v}_t^m. \tag{5}$$

Here $\eta > 0$ specifies a relative norm budget and $\epsilon > 0$ stabilizes division. The feedback scale γ_m starts at zero. In M3ED, $\zeta_t^m \in (0, 1)$ is a learned attenuation based on the update's angle and relative magnitude; the other two dataset configurations use $\zeta_t^m = 1$. Consequently, $\|\widetilde{z}_t^m - z_t^m\|_2 \leq \eta \|z_t^m\|_2$. When $g_t = 0$, the current token is unchanged.

SABER evaluates $\widetilde{z}_t^m$. In M3ED, history enters the joint predictor through feedback; admission-state statistics additionally condition the scorer. IEMOCAP and EmotionTalk also admit historical tokens to joint attention, masked by the same g_t.

3.3. History-conditioned contribution routing

SABER estimates the marginal predictive contribution of a modality after historical updating. Its attribution compares modality subsets under the *same updated representations, history admission and routing state*. This fixes the contextual decision during contribution estimation: only modality availability changes in each subset comparison.

Fixed-state Shapley targets. During training, the joint predictor evaluates the seven nonempty subsets $S \subseteq \mathcal{M}$. If $p_{t,y_t}(S)$ is the probability assigned to the ground-truth class under subset S, utility is

$$U_t(S) = \log p_{t,y_t}(S), \qquad U_t(\varnothing) = 0. \tag{6}$$

Every evaluation reuses the embeddings and admission decision from the same forward pass, together with detached routing weights. Exact Shapley valuation [26, 33] then gives

$$\phi_{t,m} = \sum_{S \subseteq \mathcal{M} \setminus \{m\}} \frac{|S|!(K - |S| - 1)!}{K!} \times \big[U_t(S \cup \{m\}) - U_t(S)\big], \tag{7}$$

where $K = |\mathcal{M}| = 3$ and S ranges over subsets of the other modalities. The sample-dependent attribution $\phi_{t,m}$ is standardized:

$$\widehat{\phi}_{t,m} = \frac{\phi_{t,m} - \bar{\phi}_t}{\max(s_{\phi,t}, \epsilon_\phi)}. \tag{8}$$

Here $\bar{\phi}_t$ and $s_{\phi,t}$ are the mean and sample standard deviation (divisor $K - 1$) across the K modalities; $\epsilon_\phi > 0$ stabilizes division. These targets supervise the scorer's relative contribution estimates within each utterance.

Bounded routing at inference. A scorer predicts $\mu_{t,m}$ from detached updated tokens, auxiliary unimodal probabilities, confidence and agreement statistics, and the admission state. At inference, this single scorer pass replaces subset evaluation. We map its centered scores to routing weights:

$$\begin{aligned} \delta_{t,m} &= \mu_{t,m} - K^{-1} \sum_{n \in \mathcal{M}} \mu_{t,n}, \\ \bar{w}_{t,m} &= \frac{1}{K}\left(1 + \lambda_s \frac{\delta_{t,m}}{\max(\max_{n \in \mathcal{M}} |\delta_{t,n}|, \epsilon)}\right), \\ w_{t,m} &= \frac{[\bar{w}_{t,m}]_+}{\sum_{n \in \mathcal{M}} [\bar{w}_{t,n}]_+}. \end{aligned} \tag{9}$$

Here $[x]_+ = \max(x, 0)$ and $\lambda_s \geq 0$ controls the deviation from uniform weighting. Equal scores yield equal weights. Centering and bounded rescaling regulate score contrast before nonnegative normalization.

For M3ED, the joint input contains the current tokens

$$Z_t = [\mathrm{sg}(w_{t,m}) \widetilde{z}_t^m]_{m \in \mathcal{M}}, \tag{10}$$

where sg is stop-gradient. A Transformer [32] processes Z_t with modality-availability masks. Let u_t^m be its current-token outputs and $M_{t,m}$ the corresponding availability indicators. The pooled state is

$$o_t = \frac{\sum_{m \in \mathcal{M}} M_{t,m} \,\mathrm{sg}(w_{t,m}) u_t^m}{\max(\sum_{m \in \mathcal{M}} M_{t,m} \,\mathrm{sg}(w_{t,m}), \epsilon)}. \tag{11}$$

SABER therefore controls both token entry and current-output pooling in this configuration. The other dataset configurations additionally admit historical tokens and use masked mean pooling over current outputs.

A linear classifier produces contextual logits ℓ_t^H and native probabilities $p_t^0 = \mathrm{softmax}(\ell_t^H)$. The selected M3ED output applies the frozen mapping

$$p_t^H = \mathrm{softmax}\left(\frac{\ell_t^H + \Delta_t + \bar{b}}{T_H}\right), \tag{12}$$

where $\Delta_t \in \mathbb{R}^C$ is a bounded residual from auxiliary unimodal evidence, $\bar{b} \in \mathbb{R}^C$ is a centered class bias, and $T_H > 0$ is a shared temperature. These fitted transformations retain their fixed parameters. For IEMOCAP and EmotionTalk, $p_t^H = p_t^0$.

3.4. Prior-relative nonverbal prediction correction

Routing reallocates evidence already present in the contextual predictor. cRBEF supplies a complementary prediction-level path through a separately trained, text-free AV expert. A TAV predictor and this AV expert produce p_t^{TAV} and p_t^{AV}, respectively; both are distinct from p_t^H. The AV expert learns audio-visual emotion cues without text input. Its posterior is compared with the training class prior $\pi \in \mathbb{R}^C$:

$$e_{t,c} = \log(p_{t,c}^{\mathrm{AV}} + \epsilon) - \log(\pi_c + \epsilon). \tag{13}$$

This class-wise log ratio specifies the correction direction relative to class prevalence. It is zero when the expert posterior matches the prior.

An utterance-level gate regulates the strength of the evidence vector:

$$\begin{aligned} \rho_t &= a_t^{\mathrm{AV}} \sigma(v^\top s_t + b_\rho), \\ p_t^E &= \mathrm{softmax}\big(\log(p_t^{\mathrm{TAV}} + \epsilon) + \rho_t e_t\big). \end{aligned} \tag{14}$$

Here a_t^{AV} indicates valid AV observations required by the expert, $v \in \mathbb{R}^6$ and b_ρ are gate parameters, and $s_t \in \mathbb{R}^6$ contains the TAV maximum probability, TAV top-two margin, AV maximum probability, maximum prior-relative evidence, top-class agreement, and TAV entropy. A single ρ_t scales the whole correction; the components of e_t determine its class-specific effect. When AV is unavailable, $\rho_t = 0$.

The corrected posterior and contextual prediction, aligned by utterance identifiers and class order, are combined as

$$p_t^R = (1 - \alpha) p_t^H + \alpha p_t^E, \qquad \alpha \in [0, 1]. \tag{15}$$

The protocol-fixed α balances complete predictions; the sample-dependent ρ_t controls AV correction within p_t^E. Thus prediction correction complements the representation-level routing.

3.5. Learning and inference

The contextual predictor is trained with

$$\begin{aligned} \mathcal{L}_{\mathrm{ctx}} = \mathcal{L}_{\mathrm{task}} &+ \lambda_u s(k) \mathcal{L}_{\mathrm{utility}} \\ &+ \lambda_{\mathrm{sub}} \mathcal{L}_{\mathrm{subset}} + \lambda_h \mathcal{L}_{\mathrm{admission}}. \end{aligned} \tag{16}$$

Task loss supervises p_t^0; utility loss regresses $\widehat{\phi}_{t,m}$; subset loss supervises the six proper nonempty modality subsets. Nonnegative coefficients weight these objectives, and $s(k)$ warms up contribution supervision over epoch k. Detached scorer inputs and deployed routing weights separate contribution regression from the task gradients used to learn representations.

Admission supervision uses out-of-fold training teachers. For each training utterance, the history-on and history-off teacher losses define $d_t = \mathrm{CE}_{\mathrm{off},t} - \mathrm{CE}_{\mathrm{on},t}$, with target $q_t^\star = \sigma(d_t/\tau_u)$ and $\tau_u > 0$. This training target is distinct from the inference estimate q_t. Admission training progresses from available history to soft gates and then straight-through hard gates. The TAV and AV experts are trained separately and frozen before fitting the cRBEF gate. Disjoint development partitions support gate fitting and configuration selection.

Inference requires no labels, teacher predictions or subset enumeration. M3ED and IEMOCAP predict $\widehat{y}_t = \arg\max_c p_{t,c}^R$. The recorded EmotionTalk configuration additionally mixes a temperature-transformed p_t^R with a separate TAV-head posterior p_t^J:

$$p_t^{\mathrm{ET}} = \omega_{\mathrm{ET}} \operatorname{softmax}\left(\log p_t^R / T_{\mathrm{ET}}\right) + (1 - \omega_{\mathrm{ET}}) p_t^J . \tag{17}$$

The reported configuration fixes $T_{\mathrm{ET}} = 3$ and $\omega_{\mathrm{ET}} = 0.85$ according to its evaluation protocol.

4. Experiments

4.1. Experimental setup

Data and evaluation. We use M3ED [38], IEMOCAP [3] and EmotionTalk [25] for evaluation. M3ED used the official television-source split. IEMOCAP held out Session 5, with happy and excited kept separate. EmotionTalk used G01/G12 for development and G03/G15 for testing. We report support-weighted F1 (WF1) and accuracy (ACC) on a percentage scale (%). Performance gains denote differences between scores on this scale. M3ED and EmotionTalk report individual runs; IEMOCAP reports the mean of three runs.

Implementation. Contextual training used AdamW. The predictor used two joint Transformer layers, six attention heads and at most three completed preceding utterances. The TAV predictor and text-free AV expert were trained separately and frozen before fitting the correction gate. Disjoint development partitions supported gate fitting and configuration selection. Dataset-specific prediction compositions follow Sec. 3; their fitted transformations remained fixed at inference. No labels, teacher predictions or subset enumeration were used at inference.

4.2. Comparison with existing methods

Tables 1–3 compare ReCoMER with existing methods on the three datasets. All listed methods use text, audio and visual inputs (TAV); literature results retain their published evaluation settings.

Method	Inputs	WF1
DialogueGCN [9]	TAV	49.93
MDI [38]	TAV	50.99
MMGCN [15]	TAV	51.18
DialogueRNN [21]	TAV	51.66
MF-EDM [16]	TAV	55.05
ReCoMER (ours)	TAV	**62.07**

Table 1. M3ED results (%). The first four baselines follow [38]; other entries retain their source settings.

M3ED. ReCoMER achieved 62.07% WF1, exceeding the strongest listed baseline, MF-EDM, by 7.02% (Table 1). The recurrent and graph-based baselines model dialogue states or multimodal interactions. ReCoMER additionally makes explicit decisions about admitting history, limiting its influence and reassessing modality contributions. The component comparisons in Sec. 4.3 examine these design choices on the same dataset.

Method	Inputs	WF1
AdaIGN [31]	TAV	70.74
MKE-IGN [30]	TAV	71.93
GRPO (LLaMA2) [5]	TAV	73.04
HiRoC [19]	TAV	73.91
HRG-SSA-gen [17]	TAV	74.63
SCoPE [18]	TAV	74.68
ReCoMER (ours)	TAV	**78.53**

Table 2. IEMOCAP results (%). ReCoMER reports a three-run mean; SCoPE reports a ten-run mean.

IEMOCAP. ReCoMER achieved a three-run mean of 78.53% WF1, exceeding the listed SCoPE mean by 3.85% (Table 2). It also exceeded HiRoC's reported 73.91% by 4.62%. This comparison is particularly relevant because HiRoC combines history routing with nonverbal prediction correction. ReCoMER introduces explicit history admission, bounded historical feedback and prior-relative corrective evidence to regulate these two sources of evidence. The sample standard deviation was 0.04%, indicating little variation across the three recorded runs. This comparison extends the evaluation beyond M3ED while retaining the distinct statistical summaries specified in the table.

Method	ACC	F1
MCTN [24]	48.99	48.53
MFM [29]	60.13	52.79
MMIM [11]	65.47	63.93
GMFN [1]	66.46	63.68
MFN [36]	67.86	67.24
MulT [28]	69.41	69.55
MISA [12]	69.86	69.53
TFN [35]	70.13	70.03
Attention [32]	70.46	70.81
LMF [20]	71.23	70.84
ReCoMER (ours)	**72.52**	**72.03**

Table 3. EmotionTalk results (%) with text, audio and vision. Literature values follow [25, Table 4].

EmotionTalk. ReCoMER obtained 72.52% ACC and 72.03% WF1 (Table 3). Its ACC exceeded that of LMF, the strongest listed baseline on this metric, by 1.29%. This comparison extends the evaluation to a second Chinese dialogue corpus using the dataset-specific prediction composition described in Sec. 3.

Evaluation scope. The M3ED and EmotionTalk results describe configurations identified through test-informed exploration under the recorded evaluation protocols.

4.3. Ablation studies

We examine the design choices behind history admission, bounded feedback, modality routing and prediction correction. Tables 4–7 report paired WF1 differences on M3ED, computed before rounding and denoted by ΔWF1 (%). The NCHA, FCCR and SABER module rows and all internal rows report selected maximal conditional gains. Settings may differ across rows; these gains are neither additive nor deletions from one common complete model.

Component	ΔWF1 (%)
Whole module	**+2.913**
Second-pass null (U)	+0.468
Second-pass null (B)	+0.447
Hard vs soft admission	+0.688
Selective vs always-on admission	+0.366
Admission supervision	+1.266

Table 4. NCHA ablations on M3ED. The module row retains SABER. U/B: unclipped/relaxed finite-cap feedback. Null pairs share admission; selective admission retains null competition.

NCHA. The NCHA comparisons distinguish null competition from the decision to apply historical feedback (Table 4). The whole module yielded a 2.913% conditional gain. Second-pass null competition improved WF1 by 0.468% and 0.447% under the two feedback settings. Selective admission improved over always-on history by 0.366%, while hard admission and admission supervision yielded 0.688% and 1.266%, respectively. These results favor both a null alternative during aggregation and an explicit decision about whether the resulting history should affect the current prediction.

Component	ΔWF1 (%)
Whole module	**+3.290**
Learned feedback mapping	+1.651
Relative-norm limit	+0.298

Table 5. FCCR ablations on M3ED. The module row retains SABER. Mapping compares learned and identity feedback without clipping; the norm limit retains learned mapping.

FCCR. FCCR examines how historical feedback is transformed and how strongly it changes the current representation (Table 5). The whole module yielded a 3.290% conditional gain. Learned feedback mapping improved WF1 by 1.651%, supporting adaptation of history to each modality. The relative-norm limit yielded a 0.298% gain in its paired comparison. This result favors controlling feedback strength relative to the current representation, in addition to learning its transformation.

Component	ΔWF1 (%)
Whole module	**+1.902**
Post- vs pre-update scoring	+0.464
Shapley supervision	+0.856
Subset supervision	+0.819
Tighter vs relaxed routing bounds	+2.019

Table 6. SABER ablations on M3ED. The module row retains FCCR. Shapley supervision is measured without subset loss; subset supervision retains Shapley supervision.

SABER. The SABER comparisons examine when contributions are estimated, how the scorer is supervised and how its weights are constrained (Table 6). The whole module yielded a 1.902% conditional gain. Scoring after historical feedback improved WF1 by 0.464%, favoring contribution estimates based on the representations actually used for prediction. Shapley and subset supervision yielded 0.856% and 0.819%, respectively. Tighter routing bounds yielded 2.019%, supporting constraints on how estimated contributions affect modality allocation at token entry and output pooling.

Component	ΔWF1 (%)
Whole module	**+7.478**
Prior-relative AV evidence	+0.556
Adaptive vs fixed-unit gate	+2.596
Arithmetic vs geometric fusion	+1.033

Table 7. cRBEF ablations on M3ED. Evidence compares prior-relative and log-posteriors with gating retained. Fusion uses identical predictions and mixing weight; module gain is over contextual prediction.

cRBEF. cRBEF improved contextual WF1 by 7.478% (Table 7). Prior-relative evidence and adaptive gating yielded 0.556% and 2.596%, respectively. These comparisons support defining the correction direction relative to the class prior and adapting its strength to each utterance. Arithmetic fusion gained 1.033% over geometric fusion, favoring arithmetic combination of corrected and contextual predictions.

Final configuration. We retained complete ReCoMER at 62.07% WF1. It preserves the admission–feedback– attribution sequence and combines it with prior-relative nonverbal correction. Its WF1 minus the corresponding NCHA, FCCR and SABER removals was -0.178%, -0.061% and -0.167%, respectively. Thus, the positive conditional gains above do not establish that every component improves the final mixture individually.

Efficiency. ReCoMER achieved 62.07% WF1 on M3ED with 13.121M parameters. FCCR, SABER and NCHA jointly added only 0.086M parameters (0.66%) over the otherwise identical configuration with these modules disabled. For published scale references, HiRoC [19] reports 17.67M trainable parameters on IEMOCAP, while the DF-GCN benchmark [41, Table 3] lists 78.59M parameters for DER-GCN in its reported configuration. Including cRBEF, the full model required 1.401 ms per utterance on an RTX 5090. Timing used dialogue batching and the median of five post-warmup runs on eight development dialogues (187 utterances), including expert computation and excluding feature extraction, data transfers and model loading.

5. Conclusion

ReCoMER integrates selective history admission, bounded contextual feedback and contribution-aware routing with complementary nonverbal correction through cRBEF. It achieves 62.066%, 78.53% and 72.03% WF1 on M3ED, IEMOCAP and EmotionTalk, respectively. Matched comparisons reveal condition-dependent effects of historical admission, feedback and modality routing, while cRBEF supplies complementary prediction-level evidence.

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